Ratios, Rates & Data

ANSWER KEY

Ratios and proportions

1 Simplify the ratio $18:24$ to lowest terms.

Divide both terms by 6: $3:4$.

2 Solve the proportion $\frac{5}{8} = \frac{x}{40}$.

Cross-multiply: $8x = 200$, so $x = 25$.

3 A recipe requires flour and sugar in the ratio $3:2$. If 12 cups of flour are used, how much sugar is needed?

Sugar $= \frac{2}{3} \times 12 = 8$ cups.

4 If $a:b = 4:7$ and $b = 35$, find a.

$\frac{a}{4} = \frac{35}{7} = 5$, so $a = 20$.

Unit rates

5 A car uses 8 gallons of gas to travel 224 miles. Find its fuel efficiency in miles per gallon.

$\frac{224}{8} = 28$ miles per gallon.

6 A printer produces 150 pages in 6 minutes. Find its rate in pages per minute, and use it to find how long it takes to print 400 pages.

Rate $= \frac{150}{6} = 25$ pages per minute. Time $= \frac{400}{25} = 16$ minutes.

Percentage change and applications

7 A price increases from 60 to 75 dollars. Find the percentage increase.

$\frac{75-60}{60} \times 100 = 25\%$.

8 A jacket originally 120 dollars is discounted 30 percent. Find the sale price.

$120 \times 0.7 = 84$ dollars.

9 A population decreases by 15 percent one year, then increases by 15 percent the next. Is the final population the same as the original?

Start with 100. After a 15\% decrease: 85. After a 15\% increase: $85 \times 1.15 = 97.75$. No — the final population is 2.25\% less than the original.

10 An investment of 2000 dollars grows to 2300 dollars after one year. Find the percentage increase.

$\frac{2300-2000}{2000} \times 100 = 15\%$.

Weighted averages

11 A student scores 80 on a test worth 40 percent of the grade and 90 on a test worth 60 percent of the grade. Find the weighted average grade.

$0.4(80) + 0.6(90) = 32 + 54 = 86$.

12 A store sells 3 items at 10 dollars each and 7 items at 20 dollars each. Find the average price per item sold.

Total revenue $= 3(10) + 7(20) = 30 + 140 = 170$ dollars. Total items $= 10$. Average $= \frac{170}{10} = 17$ dollars.

Scale and similar figures

13 A model car is built at a scale of $1 : 24$. If the actual car is 4.8 meters (480 cm) long, find the length of the model, in centimeters.

$\frac{480}{24} = 20$ centimeters.

14 A map has a scale of 1 cm to 5 km. If two cities are 8.5 cm apart on the map, find the actual distance between them.

$8.5 \times 5 = 42.5$ km.

Interpreting data tables

15 A table shows the hours worked by 4 employees in a week: A: 32, B: 40, C: 28, D: 36. Find the total hours worked and the mean hours per employee.

Total $= 32 + 40 + 28 + 36 = 136$. Mean $= \frac{136}{4} = 34$ hours.

16 A table shows quarterly revenue, in thousands of dollars: Q1: 50, Q2: 65, Q3: 70, Q4: 55. Find the percentage of annual revenue earned in Q3, to the nearest tenth of a percent.

Total $= 50 + 65 + 70 + 55 = 240$. Q3 percentage $= \frac{70}{240} \times 100 \approx 29.2\%$.

Bar graphs and data displays

17 The bar graph shows a company's monthly sales, in thousands of dollars. Find the percentage increase in sales from January to March.

$\frac{65 - 40}{40} \times 100 = 62.5\%$.

Proportional reasoning in context

18 A factory produces 450 units using 6 machines running for 8 hours. Assuming the same rate per machine-hour, how many units would 9 machines produce in 8 hours?

Rate $= \frac{450}{6 \times 8} = \frac{450}{48} = 9.375$ units per machine-hour. New machine-hours $= 9 \times 8 = 72$. Units $= 72 \times 9.375 = 675$.

19 If 5 workers can build a fence in 12 days, how many days would it take 4 workers, assuming the same work rate per worker?

Total work $= 5 \times 12 = 60$ worker-days. Days for 4 workers $= \frac{60}{4} = 15$ days.

A well-designed word problem: customer growth for a small business

20 This question has three parts. In the first quarter, a business had 240 customers, and 180 of them made a repeat purchase. (a) Find the repeat-purchase rate as a percentage. (b) In the second quarter, the number of customers grew by 25 percent, and the repeat-purchase rate held steady at the percentage found in part (a). Find the number of repeat customers in the second quarter. (c) The business sets a goal to raise the repeat-purchase rate to 80 percent by the third quarter, while keeping the same total number of customers as the second quarter. Find how many additional repeat customers (beyond part (b)'s total) are needed to reach this goal.

(a) $\frac{180}{240} \times 100 = 75\%$. (b) New customers $= 240 \times 1.25 = 300$. Repeat customers $= 0.75 \times 300 = 225$. (c) 80\% of 300 is $0.8 \times 300 = 240$ repeat customers needed. Additional customers needed $= 240 - 225 = 15$.